

Experiment Number: 4

Experiment Name: AC/DC Conversion Circuits

Station Number: 1

Submission Date: October 14, 2025

Objective

The objective of this experiment was to observe the effects of previously studied circuit elements such as diodes and transistors in the conversion of DC and AC circuits.

Equipment

- (1) Computer with PSPICE software

Procedure

Part 1 – AC to DC Conversion

1. Open the PSPICE software by searching “Capture” in the Windows menu and make a new file titled “Experiment 4.”
2. Create the circuit of figure 1 below by dropping parts, connecting with wires, and assigning values:

***Note:**

Input voltage: “Vsin”, frequency = 100Hz, $V_p = 5$ V.

Diode: D1N4001

R: 1 k Ω

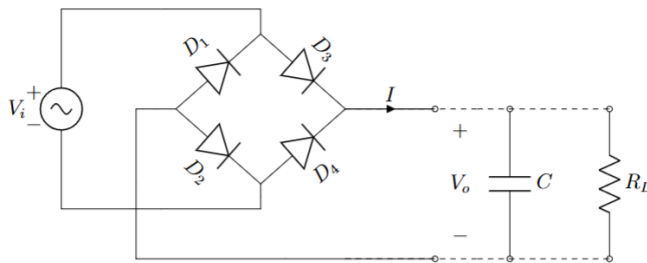


Figure 1: This is a full rectifier circuit where the voltage across the capacitor is the output voltage in DC.

3. Assign the value of the capacitor C as 0 μ F.

4. Make a new transient simulation profile and assign differential voltage markers across the source and output voltage.
5. Observe the waveform and note any differences between the input and output.
6. Repeat steps 3-5 with a capacitance value $10\ \mu\text{F}$.

Part 2 – DC to DC Conversion

1. Using the same circuit construction techniques above, construct the circuit of figure 2 below:

***Note:**

Mosfet: IRFZ24n

Diode: 1N4002

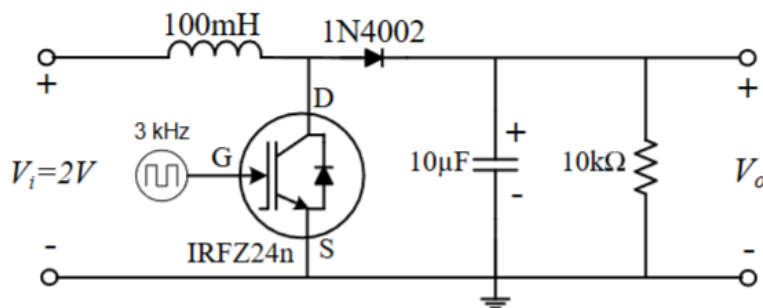


Figure 2: The following transistor circuit will be used for a DC step up conversion.

2. Assign the an input voltage “Vpulse” to the gate pin of the Mosfet with an amplitude of 10 V and a frequency of 3kHz.
3. Make a new simulation profile and assign differential voltage markers across output voltage. Generate and observe the waveforms for the following combinations:
 - $R = 1\ \text{k}\Omega$, $C = 10\ \mu\text{F}$
 - $R = 10\ \text{k}\Omega$, $C = 10\ \mu\text{F}$
 - $R = 10\ \text{k}\Omega$, $C = 100\ \mu\text{F}$

$$R = 100 \text{ k}\Omega, C = 100 \text{ }\mu\text{F}$$

4. Compare and note any differences.

Part 3 – DC to AC Conversions

1. Using the same circuit construction techniques as before, construct the circuit of figure 3 below:

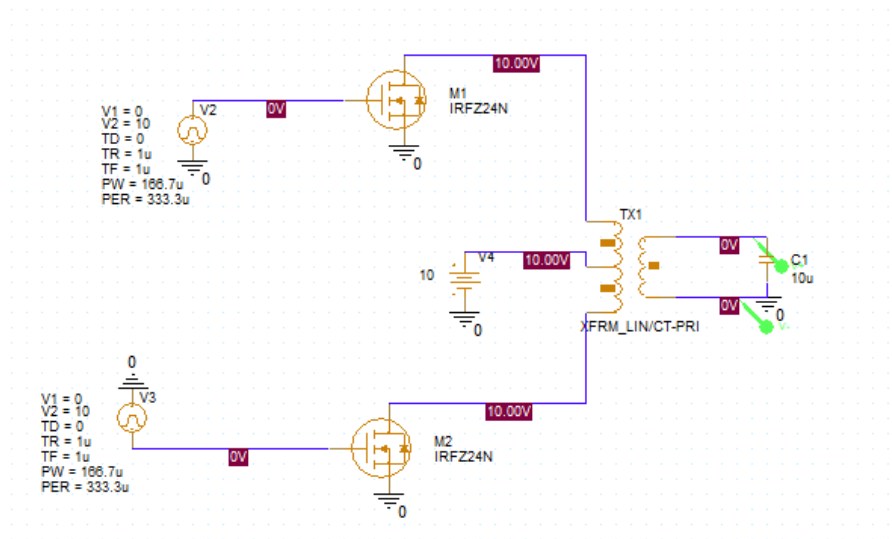


Figure 3: The following is a mosfet circuit used for DC to AC conversion.

2. Make a new simulation profile and assign differential voltage markers across the output of the transformer.
3. Observe the waveform and record anything noteworthy.

Results

Part 1 Results

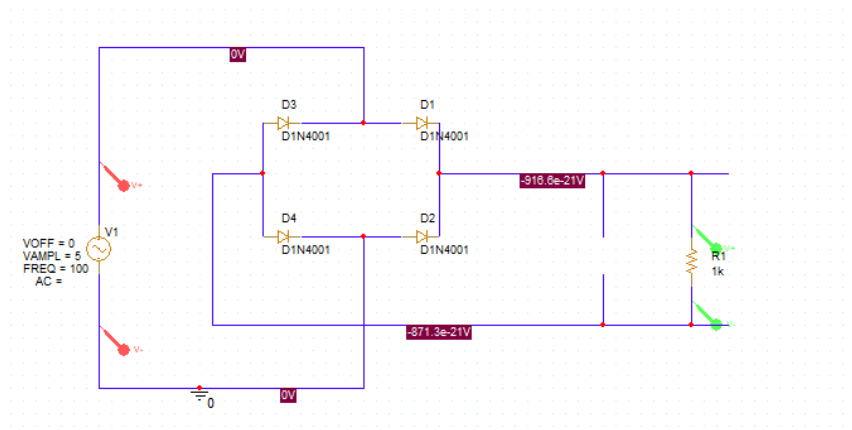


Figure 4: The following is the simulated circuit of figure 1 with a capacitance value of 0 μF .

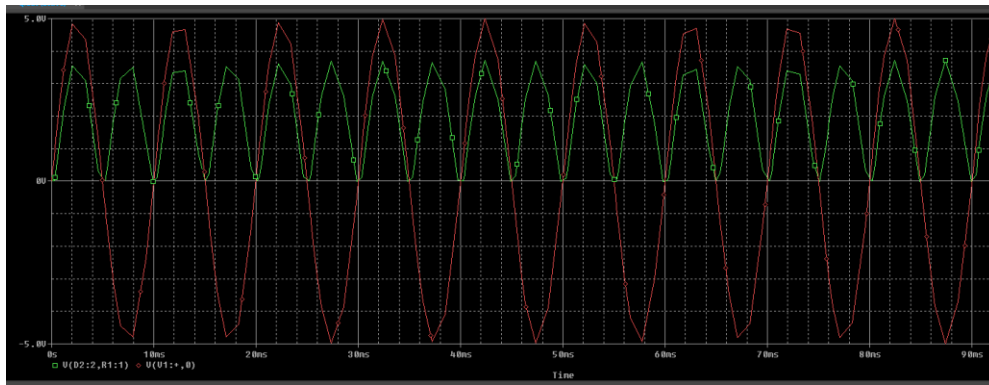


Figure 5: The following is the waveform of figure 4 with the input voltage in red and output voltage in green.

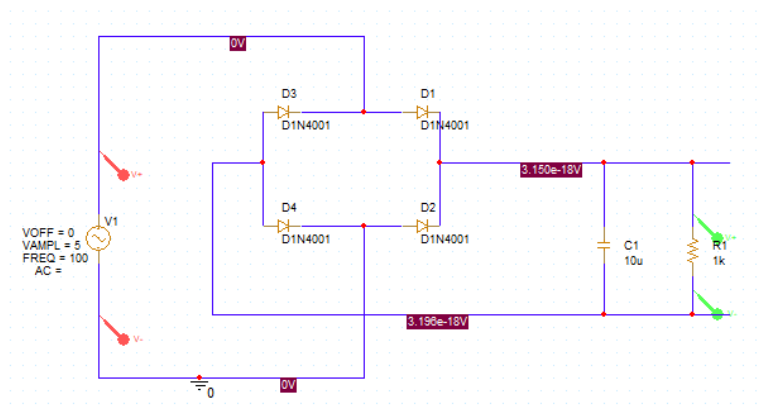


Figure 6: The following is the simulated circuit of figure 1 with a capacitance value of 10 μF .

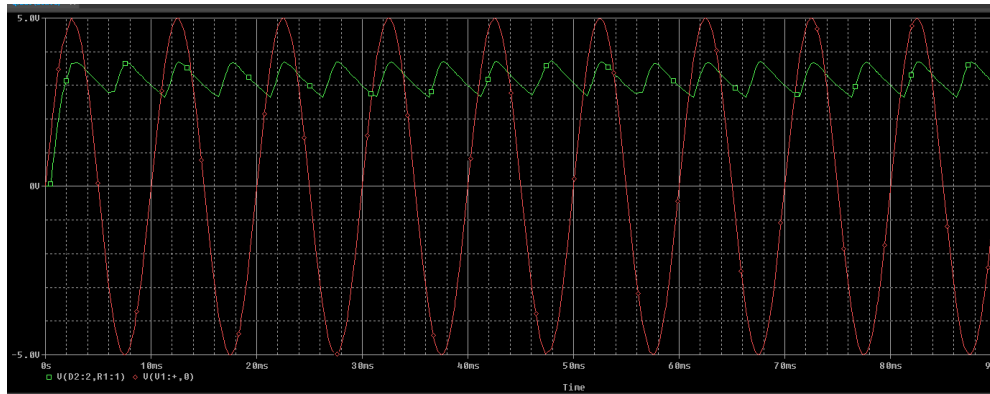


Figure 7: The following is the waveform of figure 6 with the input voltage in red and output voltage in green.

Part 2 Results

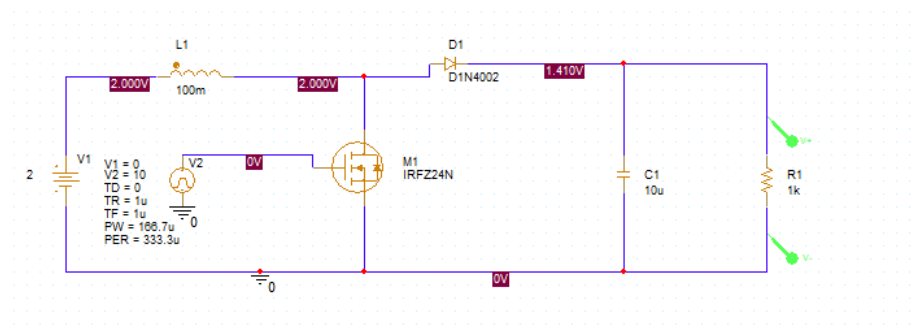


Figure 8: The following is the simulated circuit of figure 2 with a resistance value of $1k\Omega$ and capacitance value of $10\mu F$.

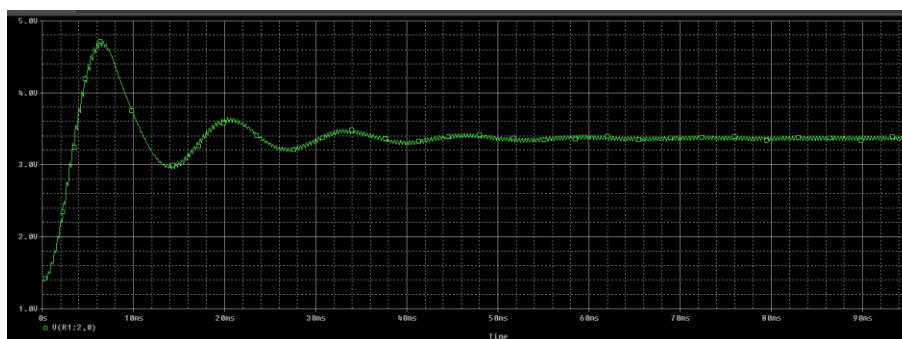


Figure 9: The following is the waveform of figure 8.

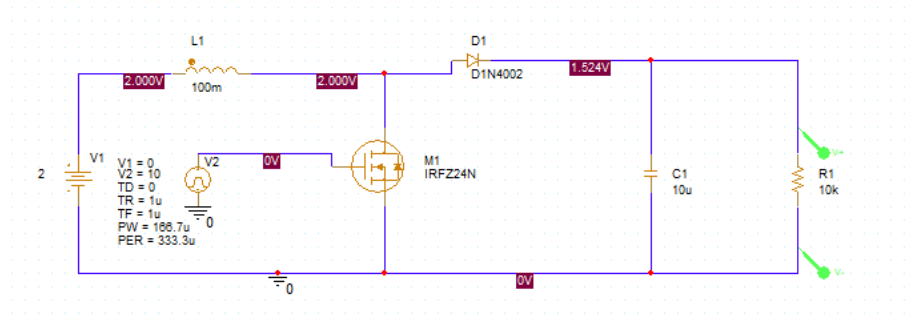


Figure 10: The following is the simulated circuit of figure 2 with a resistance value of $10\text{ k}\Omega$ and capacitance value of $10\text{ }\mu\text{F}$.

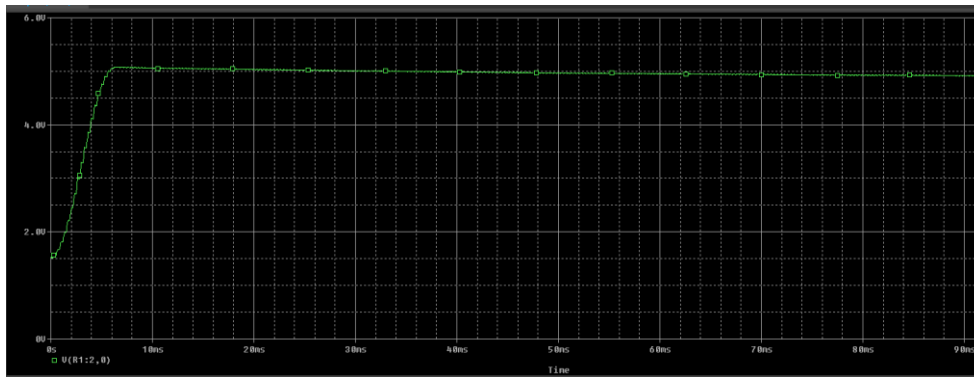


Figure 11: The following is the waveform of figure 10.

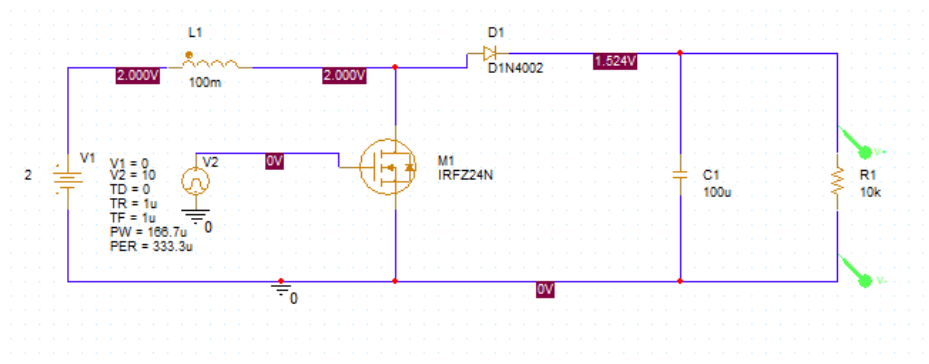


Figure 12: The following is the simulated circuit of figure 2 with a resistance value of $10\text{ k}\Omega$ and capacitance value of $100\text{ }\mu\text{F}$.

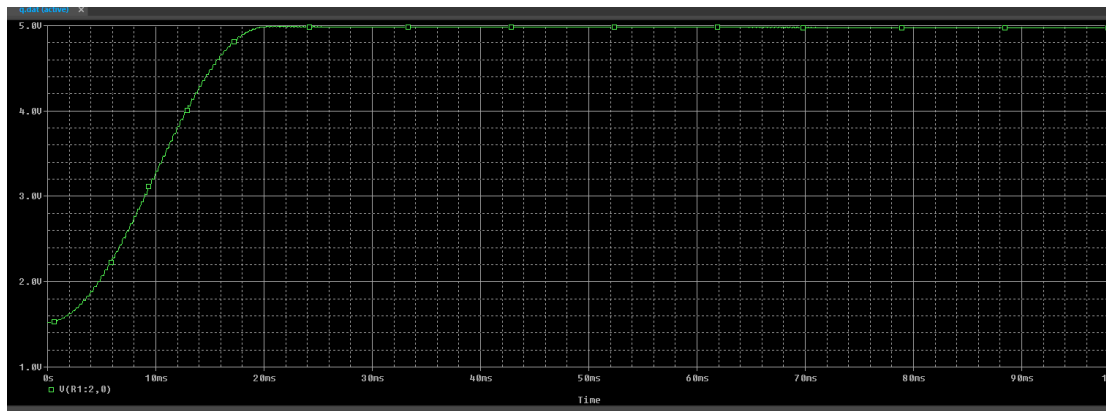


Figure 13: The following is the waveform of figure 12.

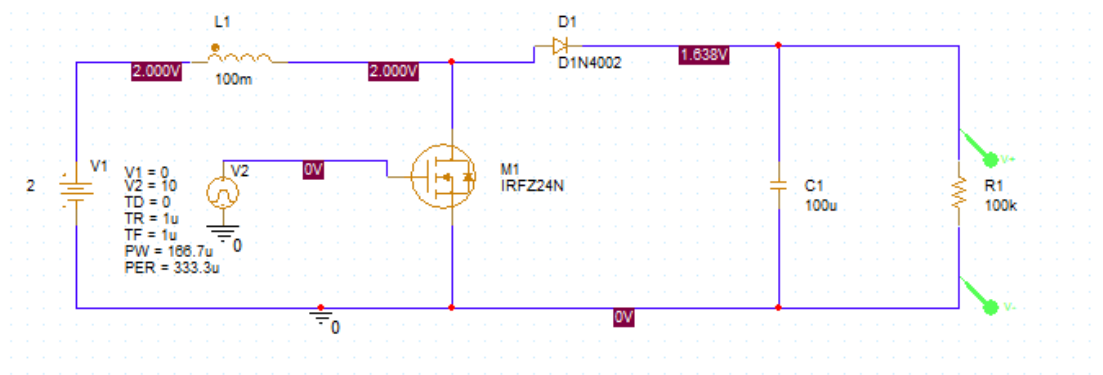


Figure 14: The following is the simulated circuit of figure 2 with a resistance value of 100 kΩ and capacitance value of 100 μF.

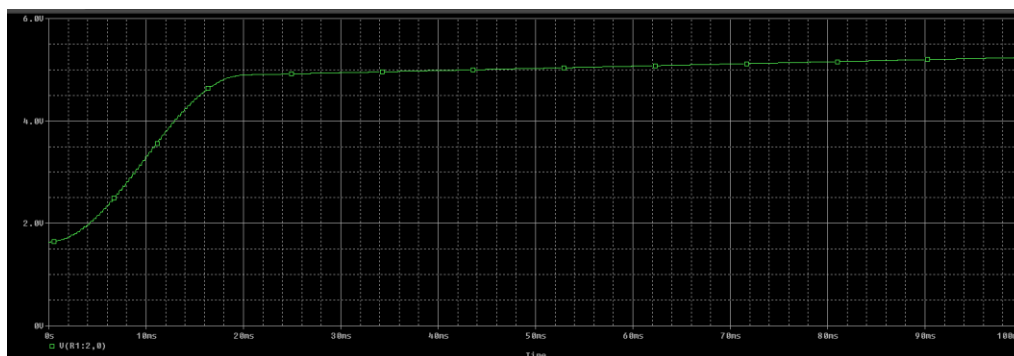


Figure 15: The following is the waveform of figure 14.

Part 3 Results

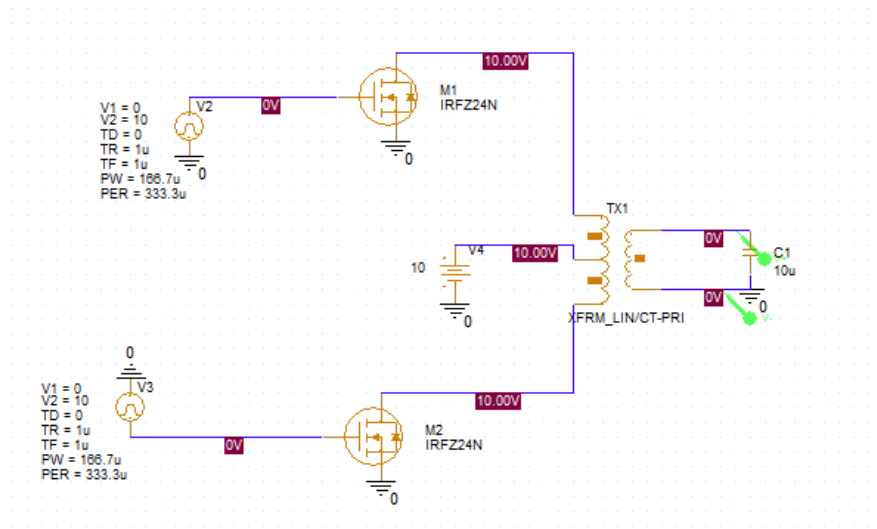


Figure 3: The following is a mosfet circuit used for DC to AC conversion. It is a simulated representation

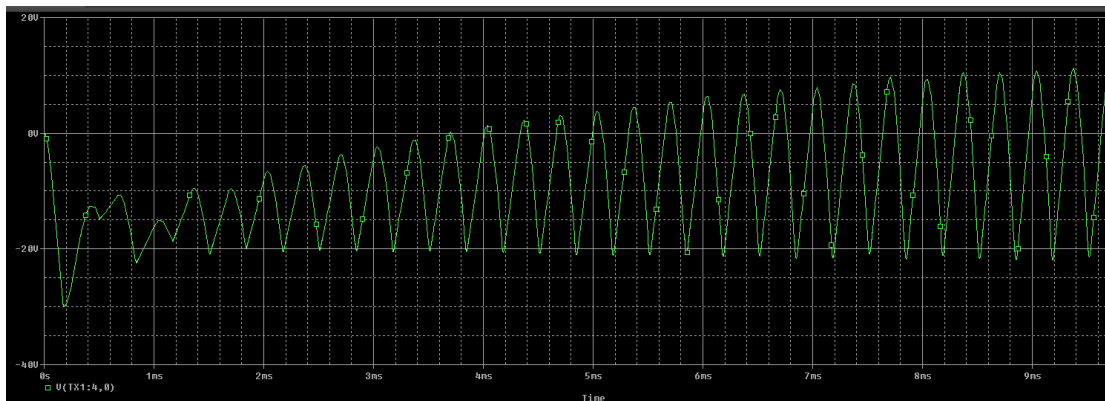


Figure 16: The following is the waveform of figure 3.

Report Questions

All plots have been given in the **Results** section of this report.

Conclusion

In the first portion of this experiment, we observed how diodes can be used to form a full bridge rectifier. As current can only flow in one direction, one would expect only the positive of a sin wave to be presented on the output, yet the implementations of two sets of diodes in this “diamond” configuration allows both sides of the signal to be captured in an overall positive waveform. We observed that with a capacitance of 0 μF , the signal remained similar to the input with a net positive inversion of the negative range of the sin wave. After increasing the capacitance to 10 μF . The signal began to resemble a true DC voltage as the charge stored in the capacitor smoothened out the drops in between each half period.

With the second portion of this experiment, we observed how mosfets and inductors could be used to step or step down a DC signal. After applying a square wave to the gate of the mosfet, the current stored in the inductor could be dumped across the diode. This resulted in a step up conversion in which the DC voltage increased from 2V to approximately 5V. However, the signal was somewhat analog with a low resistance as the waveform showed an underdamped signal. Increasing the resistance allowed for an underdamped signal which was more advantageous when producing a true DC signal at the output. Increasing the capacitance allowed for a quicker plateau which is needed as the initial logarithmic portion of the curve is not wanted for a pure DC signal.

The final portion of the circuit showed us the utilization of mosfets and transformers to produce an AC signal from a DC signal. Initially, it was difficult to understand the mode of action of the circuit elements, but after contemplation, it seems the reflected mosfets along with the switching action allowed for the production of an AC signal. However, this would only produce an analog signal, the implementation of a transformer likely acted to regulate the peak voltage.

Preparation Problems

No preparation problems were provided for this experiment.

Worksheet

No handwritten calculations were required for this experiment.